Why Cold Energy Settles

A tethered medium whose displacement is the gravitational polarization

Working note, v1.2, 2026-10-21 (v1.1 and v1.0 2026-10-20; drafted 2026-10-14). AI-assisted research programme; not peer reviewed.

Updated before deposit. The stiffness floor is raised from 5.75 pc to 24 pc by the outer-halo globular clusters (lane CFG820), and the Solar System and wide-binary numbers that scale with L_* are updated. The supply rule "each halo settles only its own supply", excluded by isolated-galaxy lensing (CFG801), is replaced by catchment supply, and the cosmic-shear result is updated (CFG650; native-input rerun CFG654). The hot-gas test result is added (CFG815).

The question. Galaxy rotation curves follow a single relation between observed and baryonic acceleration. In the framework studied here the acceleration scale is set by dark energy:

$$a_0 = \kappa c\sqrt{G\rho_{\rm DE}} \simeq 0.173\, cH_\infty, \qquad \kappa = \tfrac{1}{2}\ \text{(fitted)}.$$

Clusters, the Bullet Cluster, Local Group timing and the cosmic web also require a cold, collisionless mass component, "cold energy", with $\Omega_c/\Omega_b = 5.364$. Why should that component arrange itself, galaxy by galaxy, into exactly the profile the acceleration relation demands? This note gives the mechanism in its simplest form.

1. The medium and its displacement. Treat cold energy as a cold medium of real rest mass. Let $\mathbf{P}$ be its displacement field:

$$\partial_t \mathbf{P} = -\mathbf{j} \quad\Longrightarrow\quad \nabla\cdot\mathbf{P} = \rho - \bar\rho.$$

So the "polarization" of the gravitational vacuum is not an extra field: it is how far the medium has moved. This is a gravitational analogue of a dielectric, close in spirit to Blanchet's dipolar dark matter [2] and to superfluid dark matter [3].

2. The action. With $K = 4\pi G$ and $u = K|\mathbf{P}|$,

$$\mathcal{L} = -\frac{|\nabla\Phi - K\mathbf{P}|^2}{2K} - \rho_b\Phi - W_{\rm rem}(u) - \frac{\gamma}{2}|\nabla\mathbf{P}|^2, \qquad W_{\rm rem} = W_{\rm full} - \frac{u^2}{2K}, \qquad K\frac{dW_{\rm full}}{du} = \frac{a_0 u}{a_0 - u}.$$

The vacuum term saturates at the wall $u \to a_0$. (The subtraction in $W_{\rm rem}$ is required; pairing the completed square with $W_{\rm full}$ double-counts the quadratic term.) The stiffness $\gamma = KL_*^2/2$ is the one extra constant, $L_* \in [24\,\text{pc}, \gtrsim 1\,\text{kpc}]$. The floor comes from the outer-halo globular clusters NGC 2419, Pal 4 and Pal 14, which are Newtonian and so must be screened (95% bound; it was 5.75 pc from Saturn's ephemeris). It is not optional: without it the Sun's own field carries the simple function's constant a_0 tail, which the planetary ephemerides exclude by $\sim 2600\times$. It is irrelevant on galaxy scales.

3. The tether. Varying Φ gives the Newtonian flux of the baryons, $g - u = g_N$. Varying $\mathbf{P}$ gives the balance of the medium. Its self-gravity cancels against the quadratic part of the field energy, so each element feels only

$$\underbrace{g_N}_{\text{baryons pull in}} \qquad \text{against} \qquad \underbrace{R(u) = K\frac{dW_{\rm rem}}{du} = \frac{u^2}{a_0 - u}}_{\text{vacuum pushes back}}.$$

Like a mass on a spring, the medium comes to rest where the two balance. The balance point is unique (the energy is convex) and gives

$$g_N = R(u) \implies u = \frac{a_0 g}{a_0 + g} \implies \boxed{g_N = \frac{g^2}{g + a_0}}$$

This is the "simple" interpolating function [1]: $g \to \sqrt{a_0 g_N}$ for $g_N \ll a_0$, giving $V_f^4 = GM_b a_0$; and $g \to g_N$ for $g_N \gg a_0$. We checked it numerically: $\nu = g/g_N = 100.5$ at $g_N = 10^{-4}a_0$. **Cold energy settles because the law's profile is the only state in which the medium is at rest** (except near field null points; see the correction below).

Correction (v1.1, 2026-10-20; wording fixed in v1.2, 2026-10-21). The rest state above needs the cold density $\rho_c = \bar{\rho} + \nabla\cdot\mathbf{P}$ to stay non-negative, and near field null points it does not. Between two bodies the exact static solution needs negative cold density in a thin pancake through the saddle ($|z| \lesssim 0.05\,d$, $R \lesssim 0.43\,d$, for separation d). At $R = 0.2\,d$ this is $-89\,\bar{\rho}$ for a Milky-Way–M31-like pair ($8 \times 10^{10}\,\mathrm{M}_\odot$ each, 0.77 Mpc apart) and $-33\,\bar{\rho}$ for a cluster pair ($2 \times 10^{13}\,\mathrm{M}_\odot$ each, 5 Mpc apart), on the canonical footing. A medium of real, positive mass cannot supply this, so there the cold energy must depart from the law. This is MOND's negative "phantom" density near saddle points, which a real medium cannot realise. The total column through the saddle stays positive, though: seen edge-on, the law's negative part (-0.206 in the solution's units) is outweighed by the positive bridge around it (net $+0.016$), so neither picture predicts negative convergence there. What separates them is a *dip*: under the law the column through the saddle drops almost to zero in a narrow band (about $0.1\,d$ wide along the pair axis), and a positive medium cannot dig the part of that dip that lies below zero density. **Prediction: no saddle dip.** The missing depth is 1.9 to $2.1\,\mathrm{M}_\odot\,\mathrm{pc}^{-2}$ for the Milky-Way–M31-like pair and 4.6 to $5.1\,\mathrm{M}_\odot\,\mathrm{pc}^{-2}$ for the cluster pair (range over the two footings). How far the medium itself departs from the law there is not computed. Published stacked lensing of luminous-red-galaxy pairs [5] cannot test this yet: with every choice favouring detection, the dip gives S/N below 1 at that sample's typical mass and separation, and about twelve times as many pairs are needed for 3σ. Nothing else in this note changes. Sources: lanes AG6C and AG7.

v1.2 note. v1.1 stated the prediction as "no negative lensing trough through the saddle". That was imprecise: the law's own net column through the saddle is positive, so the prediction is a missing dip, not a missing negative trough. The numbers are unchanged.

4. How it gets there. Galaxies assemble over $\gtrsim 10^9\,\mathrm{yr}$, far slower than the medium's response time. The medium therefore follows the baryons adiabatically, and essentially no energy is radiated ($\leq 10^{-195}$ of the settling energy). A sudden start would instead relax by emitting longitudinal waves of $\mathbf{P}$, but that case is not realised in nature. The tether's projected capacity is $\bar{\Sigma} \leq a_0/\pi G$ ($\approx 214\,\mathrm{M}_\odot\,\mathrm{pc}^{-2}$). The law's own profile never reaches it.

5. What settles, and what does not.

- **Galaxies.** Nearly all baryons are condensed (stars and cold gas), and the medium tethers to them. This reproduces the radial acceleration relation: 0.100 dex on raw SPARC.
- **Solar System.** The stiff medium cannot bend on scales below L_*, so the Sun is screened even inside the Galaxy's field. Cassini's quadrupole comes out at $\leq 3.5 \times 10^{-6}$ of the 2026 bound [4] at $L_* = 5.75\,\mathrm{pc}$, even though the Galactic boost at the Sun is kept at 35–57%; it scales as L_*^{-3}, so it is $\leq 5 \times 10^{-8}$ of the bound at 24 pc. The only residue is a uniform sunward pull $GM_\odot/L_*^2 \leq 2.4 \times 10^{-16}\,\mathrm{m\,s}^{-2}$. Wide binaries come out Newtonian (-0.8σ against Gaia DR3); any boost is at most $\frac{1}{2}(s/L_*)^2$, $\leq 5 \times 10^{-5}$ at 50 kAU. Dwarf satellites much larger than L_* keep their boost.
- **Large scales.** Each system draws cold energy from its catchment, the baryons inside its turnaround radius, and only condensed baryons source the tether. With these rules the framework's own 1024^3 simulations are consistent with raw KiDS-1000 cosmic shear on both footings: $\Delta\chi^2 = -7.70$ (canonical) and -7.18 (alternative) against a same-pipeline control, with initial conditions inherited from a standard fit. Rerun with the framework's own initial conditions, the canonical footing still passes ($\Delta\chi^2 = -10.08$). The alternative-footing rerun is queued, so that footing's pass remains conditional. The earlier rule, that each halo settles only its own supply, is excluded by KiDS-1000 lensing of isolated galaxies ($\Delta\chi^2 = +197$ within 0.31 Mpc).
- **Mergers.** At collision speeds the tether cannot follow, so cold energy is collisionless. This is consistent with the Bullet Cluster.
- **Hot-gas systems (prediction).** Groups and clusters are dominated by hot, pressure-supported gas. We predict that much of their cold energy stays *untethered*: collisionless and concentrated in the core. This is a candidate for the extra core mass seen in cluster lensing ($\times 1.8$ inside $\sim$1 Mpc, 7σ). Tested since: freeing the cold share of the gas a system has expelled lowers the eRASS1 core excess from $\times 1.85$ to $\times 1.1$–1.2, but on the full 318 clusters the middle-mass third is still short by $\times 1.5$–1.6 ($Z = +4.0$ to $+4.6$). That test fails.

What is not yet derived.

- Why only condensed baryons tether. The action as written responds to all baryons.
- The energy-export channel and the frame anchor.
- The origin of the medium's mass, possibly set by ρ_{DE}.
- The relativistic completion, needed so that lensing is predicted rather than assumed.

- κ is fitted, L_* is only bounded, Ω_c/Ω_b is measured, and massive early-type galaxies still show unexplained core mass. Faint pressure-supported dwarfs sit about $+0.4\,\mathrm{dex}$ above the law below $g_{\rm bar} = 10^{-12}\,\mathrm{m\,s^{-2}}$ (PAPER55). The theory is not closed.

Sealed predictions that can kill it. (i) A Cavendish experiment whose source acceleration is changed $\times 4$ must see $|\Delta G/G| < 10^{-6}$; MOND-boosted lab models predict $\sim 3 \times 10^{-4}$. (ii) A probe beyond 1000 AU must see no non-thermal anomaly $\geq 10^{-13}\,\mathrm{m\,s^{-2}}$. (iii) A Sun–Earth saddle-point flyby sees nothing. (iv) Gaia DR4 wide binaries are Newtonian, with any boost at most $\frac{1}{2}(s/L_*)^2$.

What is new here. The interpolating function [1] and the polarization picture [2,3] are not new. New are: the scale is tied to dark energy, so a_0 is flat in redshift if $\rho_{\rm DE}$ is constant (testable with ALMA, Nov 2026); the medium-is-the-polarization identification, which makes the law the tether's unique rest state; EFE-blindness from free-fall basins (Gaia DR4, Dec 2026); and condensed-only, catchment-supplied settling that is consistent with raw cosmic shear on both footings (conditional, as stated above).

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[1] B. Famaey & J. Binney, MNRAS 363, 603 (2005). [2] L. Blanchet, Class. Quantum Grav. 24, 3529 (2007); L. Blanchet & A. Le Tiec, Phys. Rev. D 78, 024031 (2008). [3] L. Berezhiani & J. Khoury, Phys. Rev. D 92, 103510 (2015). [4] R. S. Park et al., arXiv:2602.17884 (2026). [5] S. D. Epps & M. J. Hudson, arXiv:1702.08485 (2017). Companion notes: PAPER47 (doi:10.5281/zenodo.23354043), PAPER48 (doi:10.5281/zenodo.23372002), PAPER55 (doi:10.5281/zenodo.23482936). AI assistance: Anthropic's Claude, with review by parallel AI sessions.